

CISC 699 – Applied Project

Code Submission Code Deployment Instructions

Personalized BEV energy consumption estimation framework
that integrates driver behavior with map data

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Chapter 1

Project Setup and Execution Instructions

This document describes how to run `ProjPhaseIIMain.py`, install required Python packages, and configure external dependencies such as Docker, Valhalla, ChromeDriver, and system libraries.

1.1 Quick Start

1. Navigate to the project directory:

```
cd C:\final_code\src
```

2. (Optional) Create and activate a virtual environment:

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1      # PowerShell
# or: .\.venv\Scripts\activate    # cmd.exe
```

3. Install Python dependencies:

```
pip install --upgrade pip
pip install numpy pandas scipy matplotlib requests selenium \
    joblib tensorflow scikit-learn shapely pyproj
```

Notes:

- For fewer binary issues with `pyproj` and `shapely`, consider:

```
conda install -c conda-forge pyproj shapely proj geos
```

- For CPU-only TensorFlow, install `tensorflow-cpu`.

4. Ensure external services are available:

Docker / Valhalla Routing Service

Install Docker Desktop and ensure it is running. The code expects Valhalla at `http://localhost:8002` (default).

Example deployment (Linux/Windows):

```
docker pull ghcr.io/valhalla/valhalla-scripted:latest
mkdir -p custom_files
wget -O custom_files/michigan-latest.osm.pbf \
    https://download.geofabrik.de/north-america/us/michigan-latest.osm.pbf

docker run -dt -v $PWD/custom_files:/custom_files -p 8002:8002 \
    -e build_elevation=True --name valhalla_michigan \
    ghcr.io/valhalla/valhalla-scripted:latest
```

Notes:

- Replace `$PWD` with `$(pwd)` on Windows PowerShell.
- Tile building requires significant RAM and disk space.
- Full documentation: <https://github.com/valhalla/valhalla>

Selenium / ChromeDriver

- Required by `MapDisplayHandler3`.
- Install Google Chrome.
- Download matching `chromedriver.exe` from: <https://chromedriver.chromium.org/>
- Place it on your `PATH` or in the project folder.

5. Configure runtime options (optional):

Edit `projphaseii_config.json` to adjust:

- **map_handler**: download path, timeouts, URLs
- **route_processor**: coordinate sampling, Valhalla endpoints
- **veh_model**: noise, gains, clip limits

- **inference**: LSTM method, plotting options
- **bev**: vehicle parameters for energy modeling

Missing keys fall back to built-in defaults.

6. Run the main script:

```
python ProjPhaseIIMain.py
```

Or you can also run on Windows

```
.\src\ProjPhaseIIMain.exe
```

This will:

- Load or launch a browser to obtain `route.geojson`.
- Run `RouteProcessor4` to extract node attributes from Valhalla.
- Run the PID-based vehicle simulation to generate Phase I velocity.
- Discretize the route and run LSTM inference.
- Compute BEV energy consumption and SOC evolution.

1.2 Troubleshooting

- If `pyproj` or `shapely` fail to install on Windows, use:

```
conda create -n projphaseii python=3.10
conda activate projphaseii
conda install -c conda-forge pyproj shapely proj geos
pip install numpy pandas scipy matplotlib requests selenium \
    joblib tensorflow scikit-learn
```

- TensorFlow may print informational messages at import; this is normal.
- If Selenium cannot locate `ChromeDriver`, ensure:
 - `chromedriver.exe` is on `PATH`, or
 - specify `executable_path` in the Selenium constructor.
- Running Valhalla requires downloading OSM data and building tiles; consult Valhalla documentation for advanced usage.

1.3 Files of Interest

- `projphaseii.config.json` — runtime configuration.
- `ProjPhaseIIMain.py` — main pipeline script.
- `RouteProcessor4.py` — Valhalla client and route processing.
- `PIDController2.py` — vehicle dynamics simulation.
- `inferenceLSTMUseSimSpeed.py`, `inferenceLSTM2.py` — LSTM inference utilities.
- `EnergyConsumption2.py` — BEV energy model.

1.4 Training a new Driver LSTM Model

- For training use Google Colab to speed up the training process. Upload the Jupyter notebook `LSTM_Driving.ipynb` to Google Colab and follow the instructions in the first cell of `LSTM_Driving.ipynb`.
- Deploy a Valhalla routing engine local instance for Czechia and run the `training/featureProcessor.py` on the extracted `training/ieee_vehicle_Speed_Dataset.zip` file.
- Then upload the processed `training/ieee_vehicle_Speed_Dataset.zip` folder to your Google Drive and mount it in Google Colab

1.5 Contact

For assistance, provide error messages or logs when requesting help.

Contact vasudevaraja.raju@gmail.com